Quantitative Seismology Assignment 2

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Problem 1. Properties of Green's Function Using the elastodynamic equation (2.36) satisfied by Green's function and the reciprocity theorem (2.35).

(1) Prove $\boldsymbol{G}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \boldsymbol{G}(\boldsymbol{x}, t-\boldsymbol{\tau}; \boldsymbol{\xi}, 0) = \boldsymbol{G}(\boldsymbol{x}, -\boldsymbol{\tau}; \boldsymbol{\xi}, -t)$

The elastodynamic equation (2.36) for Green's function:

$$\rho \frac{\partial^2}{\partial t^2} G_{in}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \delta_{in}\delta(\boldsymbol{x}-\boldsymbol{\xi})\delta(t-\boldsymbol{\tau}) + \frac{\partial}{\partial x_j}\left(c_{ijkl}\frac{\partial}{\partial x_l}G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau})\right) \tag{1}$$

Define the operator $\mathcal{L}_{ik}$ acting on G_{kn}:

$$\mathcal{L}_{ik} \equiv \delta_{ki}\rho\frac{\partial^2}{\partial t^2} - \frac{\partial}{\partial x_j}\left(c_{ijkl}\frac{\partial}{\partial x_l}\right) \tag{2}$$

Then (1) becomes:

$$\mathcal{L}_{ik}G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \delta_{in}\delta(\boldsymbol{x}-\boldsymbol{\xi})\delta(t-\boldsymbol{\tau}) \tag{3}$$

Shift t and $\boldsymbol{\tau}$ by Δt in (1):

$$\mathcal{L}_{ik}G_{kn}(\boldsymbol{x}, t+\Delta t; \boldsymbol{\xi}, \boldsymbol{\tau}+\Delta t) = \delta_{in}\delta(\boldsymbol{x}-\boldsymbol{\xi})\delta(t-\boldsymbol{\tau}) \tag{4}$$

Subtracting (3) from (4):

$$\mathcal{L}_{ik}\left(G_{kn}(\boldsymbol{x}, t+\Delta t; \boldsymbol{\xi}, \boldsymbol{\tau}+\Delta t) - G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau})\right) = 0 \tag{5}$$

Since $\mathcal{L}(u) = 0$ and let the Green's function of $\mathcal{L}$ be $g(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau})$, then by spatiotemporal convolution:

$$u_i(\boldsymbol{x}, t) = \int_{-\infty}^{\infty} d\tau \iiint_V g_{ij}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) f_j(\boldsymbol{\xi}, \boldsymbol{\tau})\,\mathrm{d}V(\boldsymbol{\xi}) = (g * f)_i \tag{6}$$

With $f_j = 0$, we have $u_i = (g * 0)_i = 0$. By the uniqueness theorem (see Aki & Richards, Chapter 2.3.1, strain-energy function), $u = 0$ is the only solution of $\mathcal{L}(u) = 0$. Thus from (5):

$$G_{kn}(\boldsymbol{x}, t+\Delta t; \boldsymbol{\xi}, \boldsymbol{\tau}+\Delta t) - G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = 0 \tag{7}$$

Take $\Delta t = -\boldsymbol{\tau}$ in (7):

$$G_{kn}(\boldsymbol{x}, t-\boldsymbol{\tau}; \boldsymbol{\xi}, 0) = G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) \tag{8}$$

Take $\Delta t = -t-\boldsymbol{\tau}$ in (7):

$$G_{kn}(\boldsymbol{x}, -\boldsymbol{\tau}; \boldsymbol{\xi}, -t) = G_{kn}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) \quad \text{Done (1)} \tag{9}$$

(2) Prove $G_{nm}(\boldsymbol{x}, t; \boldsymbol{\xi}, 0) = G_{mn}(\boldsymbol{\xi}, t; \boldsymbol{x}, 0)$
The reciprocity theorem (2.35):

$$\begin{aligned} &\int_{-\infty}^{\infty} \mathrm{d}t \iiint_V (u(\boldsymbol{x}, t) g(\boldsymbol{x}, \boldsymbol{\tau} - t) - v(\boldsymbol{x}, \boldsymbol{\tau} - t) f(\boldsymbol{x}, t))\, \mathrm{d}V \\ &= \int_{-\infty}^{\infty} \mathrm{d}t \iint_S (v(\boldsymbol{x}, \boldsymbol{\tau} - t) \cdot T(u(\boldsymbol{x}, t)) - u(\boldsymbol{x}, t) \cdot T(v(\boldsymbol{x}, \boldsymbol{\tau} - t)))\, \mathrm{d}S \end{aligned} \tag{10}$$

Use the homogeneous boundary condition: v, u or T equal to 0 at S.
Then (10) becomes:

$$\int_{-\infty}^{\infty} \mathrm{d}t \iiint_V u(\boldsymbol{x}, t) g(\boldsymbol{x}, \boldsymbol{\tau} - t) \mathrm{d}V = \int_{-\infty}^{\infty} \mathrm{d}t \iiint_V v(\boldsymbol{x}, \boldsymbol{\tau} - t) f(\boldsymbol{x}, t) \mathrm{d}V \tag{11}$$

Take $f(\boldsymbol{x}, t) = \delta_{in} \delta(\boldsymbol{x} - \boldsymbol{\xi}_1) \delta(t - t_1)$, $g(\boldsymbol{x}, t) = \delta_{im} \delta(\boldsymbol{x} - \boldsymbol{\xi}_2) \delta(t - t_2)$.
Then $u_i(\boldsymbol{x}, t) = G_{in}(\boldsymbol{x}, t; \boldsymbol{\xi}_1, t_1)$, $v_i(\boldsymbol{x}, t) = G_{im}(\boldsymbol{x}, t; \boldsymbol{\xi}_2, t_2)$.
Substituting into (11):

$$\begin{aligned} &\int_{-\infty}^{\infty} \mathrm{d}t \iiint_V G_{in}(\boldsymbol{x}, t; \boldsymbol{\xi}_1, t_1) \delta_{im} \delta(\boldsymbol{x} - \boldsymbol{\xi}_2) \delta(\boldsymbol{\tau} - t - t_2) \mathrm{d}V \\ &= \int_{-\infty}^{\infty} \mathrm{d}t \iiint_V G_{im}(\boldsymbol{x}, \boldsymbol{\tau} - t; \boldsymbol{\xi}_2, t_2) \delta_{in} \delta(\boldsymbol{x} - \boldsymbol{\xi}_1) \delta(t - t_1) \mathrm{d}V \end{aligned} \tag{12}$$

Evaluating the integrals in (12):

$$G_{mn}(\boldsymbol{\xi}_2, \boldsymbol{\tau} - t_2; \boldsymbol{\xi}_1, t_1) = G_{nm}(\boldsymbol{\xi}_1, \boldsymbol{\tau} - t_1; \boldsymbol{\xi}_2, t_2) \tag{13}$$

By (8), L.H.S. of (13) shift $\boldsymbol{\tau} - t_2$, t_1 by $-t_1$:

$$G_{mn}(\boldsymbol{\xi}_2, \boldsymbol{\tau} - t_2 - t_1; \boldsymbol{\xi}_1, 0) \tag{14}$$

R.H.S. of (13) shift $\boldsymbol{\tau} - t_1$, t_2 by $-t_2$:

$$G_{nm}(\boldsymbol{\xi}_1, \boldsymbol{\tau} - t_1 - t_2; \boldsymbol{\xi}_2, 0) \tag{15}$$

Let $\boldsymbol{\xi}_2 = \boldsymbol{\xi}$, $\boldsymbol{\xi}_1 = \boldsymbol{x}$, $\boldsymbol{\tau} - t_1 - t_2 = t$ in (14) and (15):

$$G_{mn}(\boldsymbol{\xi}, t; \boldsymbol{x}, 0) = G_{nm}(\boldsymbol{x}, t; \boldsymbol{\xi}, 0) \tag{16}$$

(3) Prove $G_{nm}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \boldsymbol{x}, -t)$
Replace t by $t - \boldsymbol{\tau}$ in (16):

$$G_{mn}(\boldsymbol{\xi}, t - \boldsymbol{\tau}; \boldsymbol{x}, 0) = G_{nm}(\boldsymbol{x}, t - \boldsymbol{\tau}; \boldsymbol{\xi}, 0) \tag{17}$$

L.H.S. of (17) shift $t - \boldsymbol{\tau}$, 0 by $-t$:

$$G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \boldsymbol{x}, -t) \tag{18}$$

R.H.S. of (17) shift $t - \boldsymbol{\tau}$, 0 by $\boldsymbol{\tau}$:

$$G_{nm}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) \tag{19}$$

Then from (18) and (19):

$$G_{nm}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = G_{mn}(\boldsymbol{\xi}, -\boldsymbol{\tau}; \boldsymbol{x}, -t) \tag{20}$$

Problem 2. Conservation of Total Momentum $f^{[u]}$ is body-force equivalent for displacement discontinuity across fault plane.

$[u_i(\boldsymbol{\xi},\boldsymbol{\tau})] \triangleq u_i^+(\boldsymbol{\xi},\boldsymbol{\tau}) - u_i^-(\boldsymbol{\xi},\boldsymbol{\tau})$ is displacement discontinuity across fault plane. ν is normal vector of fault plane. η is spatial coordinates. $\Sigma(\boldsymbol{\xi})$ is fault plane.

The body-force equivalent:

$$f_p^{[u]}(\eta,\boldsymbol{\tau}) = -\iint_\Sigma [u_i(\boldsymbol{\xi},\boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}\Sigma(\boldsymbol{\xi}) \tag{21}$$

(1) Prove

$$\iiint_V f^{[u]}(\eta,\boldsymbol{\tau})\mathrm{d}V(\eta) = 0 \tag{22}$$

Substituting (21) into (22):

$$\begin{aligned}\iiint_V f_p^{[u]}(\eta,\boldsymbol{\tau})\mathrm{d}V(\eta) &= \iiint_V \left(-\iint_\Sigma [u_i(\boldsymbol{\xi},\boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}\Sigma(\boldsymbol{\xi})\right)\mathrm{d}V(\eta)\\ &= -\iint_\Sigma [u_i(\boldsymbol{\xi},\boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\left(\iiint_V \frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}V(\eta)\right)\mathrm{d}\Sigma(\boldsymbol{\xi})\end{aligned} \tag{23}$$

Evaluating the volume integral:

$$\begin{aligned}\iiint_V \frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}V(\eta) &= \iiint_V \frac{\partial}{\partial\eta_q}\left(\delta(\eta_1-\xi_1)\delta(\eta_2-\xi_2)\delta(\eta_3-\xi_3)\right)\mathrm{d}\eta_q\mathrm{d}\eta_i\mathrm{d}\eta_j\\ &= \iint \delta(\eta_1-\xi_1)\delta(\eta_2-\xi_2)\delta(\eta_3-\xi_3)\Big|_{\eta_q^-}^{\eta_q^+}\mathrm{d}\eta_i\mathrm{d}\eta_j\\ &= 0\end{aligned} \tag{24}$$

The boundary term vanishes since $\delta(\eta-\boldsymbol{\xi}) = 0$ at $\eta_q \to \pm\infty$.
Substituting (24) into (23):

$$\iiint_V f^{[u]}(\eta,\boldsymbol{\tau})\mathrm{d}V(\eta) = 0 \tag{25}$$

(2) Prove $\iiint_V (\eta-\eta_0)\times f^{[u]}(\eta,\boldsymbol{\tau})\mathrm{d}V(\eta) = 0$
In component form:

$$\iiint_V (\eta-\eta_0)\times f^{[u]}(\eta,\boldsymbol{\tau})\mathrm{d}V(\eta) = \iiint_V \varepsilon_{mpn}(\eta_m-\eta_{0m})f_p^{[u]}(\eta,\boldsymbol{\tau})\boldsymbol{e}_n\mathrm{d}V(\eta) \tag{26}$$

Substituting the body-force equivalent:

$$\begin{aligned}&\iiint_V \varepsilon_{mpn}(\eta_m-\eta_{0m})f_p^{[u]}(\eta,\boldsymbol{\tau})\boldsymbol{e}_n\mathrm{d}V(\eta)\\ &= \boldsymbol{e}_n\iiint_V \varepsilon_{mpn}(\eta_m-\eta_{0m})\left(-\iint_\Sigma [u_i(\boldsymbol{\xi},\boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}\Sigma(\boldsymbol{\xi})\right)\mathrm{d}V(\eta)\\ &= -\boldsymbol{e}_n\iint_\Sigma \varepsilon_{mpn}[u_i(\boldsymbol{\xi},\boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\left(\iiint_V (\eta_m-\eta_{0m})\frac{\partial\delta(\eta-\boldsymbol{\xi})}{\partial\eta_q}\mathrm{d}V(\eta)\right)\mathrm{d}\Sigma(\boldsymbol{\xi})\end{aligned} \tag{27}$$

Evaluating the volume integral by integration by parts:

$$
\begin{aligned}
&\iiint_V (\eta_m - \eta_{0m}) \frac{\partial \delta(\eta - \boldsymbol{\xi})}{\partial \eta_q} \mathrm{d}V(\eta) \\
&= \iiint_V (\eta_m - \eta_{0m}) \frac{\partial}{\partial \eta_q} \left(\delta(\eta_1 - \xi_1)\delta(\eta_2 - \xi_2)\delta(\eta_3 - \xi_3)\right) \mathrm{d}\eta_q \mathrm{d}\eta_i \mathrm{d}\eta_j \\
&= \iint (\eta_m - \eta_{0m})\delta(\eta_1 - \xi_1)\delta(\eta_2 - \xi_2)\delta(\eta_3 - \xi_3)\Big|_{\eta_q^-}^{\eta_q^+} \mathrm{d}\eta_i \mathrm{d}\eta_j \\
&\quad - \iiint_V \delta(\eta - \boldsymbol{\xi}) \frac{\partial(\eta_m - \eta_{0m})}{\partial \eta_q} \mathrm{d}\eta_q \mathrm{d}\eta_i \mathrm{d}\eta_j \\
&= 0 - \iiint_V \delta(\eta - \boldsymbol{\xi})\delta_{mq} \mathrm{d}V(\eta) \\
&= -\delta_{mq}
\end{aligned} \tag{28}
$$

Substituting (28) into (27):

$$
-\boldsymbol{e}_n \iint_\Sigma \varepsilon_{mpn}[u_i(\boldsymbol{\xi}, \boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})(-\delta_{mq})\mathrm{d}\Sigma(\boldsymbol{\xi}) = \boldsymbol{e}_n \iint_\Sigma \varepsilon_{qpn}[u_i(\boldsymbol{\xi}, \boldsymbol{\tau})]\nu_j(\boldsymbol{\xi})c_{ijpq}(\boldsymbol{\xi})\mathrm{d}\Sigma(\boldsymbol{\xi}) \tag{29}
$$

Using the symmetry of elastic tensor $c_{ijpq} = c_{ijqp}$:

$$
\begin{aligned}
\varepsilon_{qpn}c_{ijpq} &= \varepsilon_{qpn}\frac{c_{ijpq} + c_{ijqp}}{2} = \frac{1}{2}\left(\varepsilon_{qpn}c_{ijpq} + \varepsilon_{qpn}c_{ijqp}\right) \\
&= \frac{1}{2}\left(\varepsilon_{qpn}c_{ijpq} + \varepsilon_{pqn}c_{ijpq}\right) = \frac{1}{2}\left(\varepsilon_{qpn} - \varepsilon_{qpn}\right)c_{ijpq} = 0
\end{aligned} \tag{30}
$$

Substituting (30) into (29):

$$
\iiint_V (\eta - \eta_0) \times f^{[u]}(\eta, \boldsymbol{\tau})\mathrm{d}V(\eta) = 0 \quad \text{Done (2)} \tag{31}
$$

Problem 3 Green's Function for Poisson Equation

Prove $G(\boldsymbol{x}; \boldsymbol{\xi}) = -\dfrac{1}{4\pi|\boldsymbol{x} - \boldsymbol{\xi}|}$ is Green's function of:

$$
\nabla^2 G(\boldsymbol{x}; \boldsymbol{\xi}) = \delta(\boldsymbol{x} - \boldsymbol{\xi}) \tag{32}
$$

Proof:

By Problem 4, the Green's function (38) is:

$$
g(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \frac{1}{4\pi c^2|\boldsymbol{x} - \boldsymbol{\xi}|}\delta\left(t - \boldsymbol{\tau} - \frac{|\boldsymbol{x} - \boldsymbol{\xi}|}{c}\right)
$$

which satisfies (39):

$$
\ddot{g}(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = c^2\nabla^2 g(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) + \delta(t - \boldsymbol{\tau})\delta(\boldsymbol{x} - \boldsymbol{\xi})
$$

Set the body force as:

$$
f(\boldsymbol{x}, t) = c^2\delta(\boldsymbol{x} - \boldsymbol{\xi}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} c^2\delta(\boldsymbol{x}' - \boldsymbol{\xi})\delta(\boldsymbol{x} - \boldsymbol{x}')\delta(t - t')\,\mathrm{d}t' d\boldsymbol{x}' \tag{33}
$$

Then the wave equation with this source is:

$$\ddot{u}(\boldsymbol{x},t;\boldsymbol{\xi}) = c^2\nabla^2 u(\boldsymbol{x},t;\boldsymbol{\xi}) + c^2\delta(\boldsymbol{x}-\boldsymbol{\xi}) \tag{34}$$

The solution by convolution with (38):

$$\begin{aligned} u(\boldsymbol{x},t;\boldsymbol{\xi}) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} c^2\delta(\boldsymbol{x}'-\boldsymbol{\xi})\frac{1}{4\pi c^2|\boldsymbol{x}-\boldsymbol{x}'|}\delta\left(t-t'-\frac{|\boldsymbol{x}-\boldsymbol{x}'|}{c}\right)\mathrm{d}t'd\boldsymbol{x}' \\ &= \frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|} \end{aligned} \tag{35}$$

where the result is independent of c and t.
From (34), letting $c \to \infty$:

$$\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right)u = c^2\delta(\boldsymbol{x}-\boldsymbol{\xi}) \quad \Leftrightarrow \quad \nabla^2(-u) = \delta(\boldsymbol{x}-\boldsymbol{\xi}) \tag{36}$$

Thus $-u = -\dfrac{1}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|}$ is the solution of Possion Equation (32).
Therefore:

$$G(\boldsymbol{x};\boldsymbol{\xi}) = -\frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|} \tag{37}$$

Problem 4. Green's Function for Wave Equation Prove:

$$g(\boldsymbol{x},t;\boldsymbol{\xi},\boldsymbol{\tau}) = \frac{1}{4\pi c^2|\boldsymbol{x}-\boldsymbol{\xi}|}\delta\left(t-\boldsymbol{\tau}-\frac{|\boldsymbol{x}-\boldsymbol{\xi}|}{c}\right) \tag{38}$$

is the solution of:

$$\ddot{g}(\boldsymbol{x},t;\boldsymbol{\xi},\boldsymbol{\tau}) = c^2\nabla^2 g(\boldsymbol{x},t;\boldsymbol{\xi},\boldsymbol{\tau}) + \delta(t-\boldsymbol{\tau})\delta(\boldsymbol{x}-\boldsymbol{\xi}) \tag{39}$$

Proof:
Let $\boldsymbol{G}(p;\boldsymbol{\xi}) = \mathcal{L}(g) = \int_0^\infty g(\boldsymbol{x},t;\boldsymbol{\xi},\boldsymbol{\tau})e^{-pt}\mathrm{d}t$.
Then:

$$\mathcal{L}(\ddot{g}) = p^2G(p) - pg_{t=0} - \dot{g}_{t=0} = p^2G(p) \tag{40}$$

Taking Laplace transform of (39):

$$p^2G = c^2\nabla^2 G + e^{-p\boldsymbol{\tau}}\delta(\boldsymbol{x}-\boldsymbol{\xi}) \tag{41}$$

Let $r = |\boldsymbol{x}-\boldsymbol{\xi}|$. By spherical symmetry, G is independent of θ and φ:

$$\nabla^2 G = \left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right)G = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial G}{\partial r}\right) = \frac{1}{r}\frac{\partial^2(rG)}{\partial r^2} \tag{42}$$

For $r > 0$, (41) becomes:

$$p^2G = \frac{c^2}{r}\frac{\partial^2(rG)}{\partial r^2} \Rightarrow \frac{p^2}{c^2}(rG) = \frac{\partial^2(rG)}{\partial r^2} \tag{43}$$

The general solution of (43):

$$G = \frac{A}{r}e^{\pm\frac{p}{c}r} \tag{44}$$

Since $\mathrm{Re}(p) > 0$, $c > 0$, $r > 0$, for G to be finite as $r \to \infty$:

$$G = \frac{A}{r} e^{-\frac{p}{c} r} \tag{45}$$

To determine A, take the open ball V_ε centered at $r = 0$ with radius ε:

$$\iiint_{V_\varepsilon} p^2 G \,\mathrm{d}V = c^2 \iiint_{V_\varepsilon} \nabla^2 G \,\mathrm{d}V + \iiint_{V_\varepsilon} e^{-p\boldsymbol{\tau}} \delta(r) \,\mathrm{d}V \tag{46}$$

For the first term on the L.H.S. of (46):

$$\begin{aligned}
\iiint_{V_\varepsilon} p^2 G \,\mathrm{d}V &= p^2 \int_0^{2\pi} \int_0^{\pi} \int_0^{\varepsilon} \frac{A}{r} e^{-\frac{p}{c} r} r^2 \sin\theta \,\mathrm{d}r \mathrm{d}\theta \mathrm{d}\varphi \\
&= 4\pi p^2 A \int_0^{\varepsilon} r e^{-\frac{p}{c} r} \mathrm{d}r = 4\pi p^2 A (\varepsilon - 0) \xi e^{-\frac{p}{c}\xi} \quad (\xi \in [0, \varepsilon] \text{ by mean value theorem}) \\
&= 0 \quad (\text{as } \varepsilon \to 0)
\end{aligned} \tag{47}$$

For the first term on the R.H.S. of (46), by divergence theorem:

$$c^2 \iiint_{V_\varepsilon} \nabla^2 G \,\mathrm{d}V = c^2 \oiint_{S_\varepsilon} \nabla G \cdot \hat{\boldsymbol{n}} \,\mathrm{d}S \tag{48}$$

In spherical coordinates:

$$\nabla = \hat{\boldsymbol{e}}_r \frac{\partial}{\partial r} + \hat{\boldsymbol{e}}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{e}}_\varphi \frac{1}{r \sin\theta} \frac{\partial}{\partial \varphi} \tag{49}$$

From (45):

$$\nabla G = \hat{\boldsymbol{e}}_r \frac{\partial}{\partial r} \left(\frac{A}{r} e^{-\frac{p}{c} r} \right) = \hat{\boldsymbol{e}}_r A \frac{-\frac{p}{c} r e^{-\frac{p}{c} r} - e^{-\frac{p}{c} r}}{r^2} \tag{50}$$

Evaluating (48):

$$\begin{aligned}
c^2 \oiint_{S_\varepsilon} \nabla G \cdot \hat{\boldsymbol{n}} \,\mathrm{d}S &= \int_0^{2\pi} \int_0^{\pi} c^2 \frac{A}{r^2} \left(-\frac{p}{c} r e^{-\frac{p}{c} r} - e^{-\frac{p}{c} r} \right) r^2 \sin\theta \,\mathrm{d}\theta \mathrm{d}\varphi \\
&= 4\pi c^2 A \left(-\frac{p}{c} \varepsilon e^{-\frac{p}{c}\varepsilon} - e^{-\frac{p}{c}\varepsilon} \right) \\
&= -4\pi c^2 A \quad (\text{as } \varepsilon \to 0)
\end{aligned} \tag{51}$$

For the source and second term on the R.H.S. of (46):

$$\iiint_{V_\varepsilon} e^{-p\boldsymbol{\tau}} \delta(r) \,\mathrm{d}V = e^{-p\boldsymbol{\tau}} \tag{52}$$

(51), Substituting (47) , and (52) into (46):

$$0 = -4\pi c^2 A + e^{-p\boldsymbol{\tau}} \Rightarrow A = \frac{e^{-p\boldsymbol{\tau}}}{4\pi c^2} \tag{53}$$

Therefore from (45) and (53):

$$G = \frac{1}{4\pi c^2 r} e^{-p\left(\frac{r}{c} + \boldsymbol{\tau}\right)} \tag{54}$$

By the inverse Laplace transform (Bromwich integral):

$$g = \frac{1}{2\pi i} \lim_{\delta \to \infty} \int_{s-i\delta}^{s+i\delta} G\, e^{pt} \mathrm{d}p \tag{55}$$

Since there is no pole in the right half-plane (see Figure), we can evaluate (55):

$$\begin{aligned} g &= \frac{1}{2\pi i} \lim_{\delta \to \infty} \int_{s-i\delta}^{s+i\delta} \frac{e^{-p\left(\frac{r}{c}+\boldsymbol{\tau}-t\right)}}{4\pi c^2 r} \mathrm{d}p \\ &= \frac{e^{-s\left(\frac{r}{c}+\boldsymbol{\tau}-t\right)}}{4\pi c^2 r} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\delta\left(\frac{r}{c}+\boldsymbol{\tau}-t\right)} \mathrm{d}\delta \\ &= \frac{e^{-s\left(\frac{r}{c}+\boldsymbol{\tau}-t\right)}}{4\pi c^2 r} \delta\left(t - \frac{r}{c} - \boldsymbol{\tau}\right) \end{aligned} \tag{56}$$

(by the definition of Dirac delta function, $f(x)\delta(x-a) = f(a)\delta(x-a)$)

$$= \frac{\delta\left(t - \frac{r}{c} - \boldsymbol{\tau}\right)}{4\pi c^2 r} \tag{57}$$

Therefore from (57):

$$g(\boldsymbol{x}, t; \boldsymbol{\xi}, \boldsymbol{\tau}) = \frac{\delta\left(t - \boldsymbol{\tau} - \dfrac{|\boldsymbol{x}-\boldsymbol{\xi}|}{c}\right)}{4\pi c^2 |\boldsymbol{x}-\boldsymbol{\xi}|} \quad \text{Done (4)} \tag{58}$$

is the solution of (39).

Problem 5 Helmholtz Potentials

According to Box 4.2, a given vector function $\mathbf{f}(\boldsymbol{x})$ can be decomposed into:

$$\mathbf{f}(\boldsymbol{x}) = \nabla\Phi(\boldsymbol{x}) + \nabla \times \Psi(\boldsymbol{x}) \tag{59}$$

Potentials $\Psi(\boldsymbol{x})$ and $\Phi(\boldsymbol{x})$ can be obtained by generating function $\mathbf{W}(\boldsymbol{x})$:

$$\Psi(\boldsymbol{x}) = -\nabla \times \mathbf{W}(\boldsymbol{x}) \quad \text{and} \quad \Phi(\boldsymbol{x}) = \nabla \cdot \mathbf{W}(\boldsymbol{x}) \tag{60}$$

where:

$$\mathbf{W}(\boldsymbol{x}) = -\iiint_V \frac{\mathbf{f}(\boldsymbol{\xi})}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|} \mathrm{d}V(\boldsymbol{\xi}) \tag{61}$$

(1) $\mathbf{f}(\boldsymbol{x}) = X_i(t)\boldsymbol{e}_i\delta(\boldsymbol{x})$

The generating function:

$$\mathbf{W}(\boldsymbol{x}) = -\iiint_V \frac{X_i(t)\boldsymbol{e}_i\delta(\boldsymbol{\xi})}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|} \mathrm{d}V(\boldsymbol{\xi}) = -\frac{X_i(t)\boldsymbol{e}_i}{4\pi|\boldsymbol{x}|} \tag{62}$$

Using $\dfrac{\partial|\boldsymbol{x}|}{\partial x_m} = \dfrac{x_m}{|\boldsymbol{x}|}$ and $\dfrac{\partial}{\partial x_m}\left(\dfrac{1}{|\boldsymbol{x}|}\right) = -\dfrac{x_m}{|\boldsymbol{x}|^3}$:

The scalar potential:

$$\Phi(\boldsymbol{x}) = \nabla \cdot \mathbf{W}(\boldsymbol{x}) = -\frac{\partial}{\partial x_m}\left(\frac{X_m(t)}{4\pi|\boldsymbol{x}|}\right) = -\frac{X_m(t)}{4\pi}\left(-\frac{x_m}{|\boldsymbol{x}|^3}\right) = \frac{X_m(t)x_m}{4\pi|\boldsymbol{x}|^3} \tag{63}$$

The vector potential:

$$\begin{aligned}\Psi(\boldsymbol{x}) &= -\nabla\times\mathbf{W}(\boldsymbol{x}) = \nabla\times\left(\frac{X_i(t)\boldsymbol{e}_i}{4\pi|\boldsymbol{x}|}\right) = \varepsilon_{mnp}\frac{\partial}{\partial x_m}\left(\frac{X_n(t)}{4\pi|\boldsymbol{x}|}\right)\boldsymbol{e}_p \\ &= -\varepsilon_{mnp}\frac{x_m}{|\boldsymbol{x}|^3}\frac{X_n(t)}{4\pi}\boldsymbol{e}_p = -\frac{1}{4\pi}\frac{(\boldsymbol{x}\times\mathbf{X}(t))}{|\boldsymbol{x}|^3}\end{aligned} \tag{64}$$

(2) $\mathbf{f}(\boldsymbol{x}) = -M_{ij}(t)\boldsymbol{e}_i\frac{\partial}{\partial x_j}\delta(\boldsymbol{x})$

The generating function:

$$\mathbf{W}(\boldsymbol{x}) = -\iiint_V \frac{-M_{ij}(t)\boldsymbol{e}_i\frac{\partial}{\partial\xi_j}\delta(\boldsymbol{\xi})}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|}\mathrm{d}V(\boldsymbol{\xi}) \tag{65}$$

Using integration by parts and $\frac{\partial}{\partial\xi_j}\left(\frac{1}{|\boldsymbol{x}-\boldsymbol{\xi}|}\right) = \frac{x_j-\xi_j}{|\boldsymbol{x}-\boldsymbol{\xi}|^3}$:

$$\begin{aligned}\mathbf{W}(\boldsymbol{x}) &= \boldsymbol{e}_i\frac{M_{ij}(t)}{4\pi}\iiint_V \frac{\frac{\partial}{\partial\xi_j}\delta(\boldsymbol{\xi})}{|\boldsymbol{x}-\boldsymbol{\xi}|}\mathrm{d}V(\boldsymbol{\xi}) \\ &= \boldsymbol{e}_i\frac{M_{ij}(t)}{4\pi}\left(\iint_S \frac{\delta(\boldsymbol{\xi})}{|\boldsymbol{x}-\boldsymbol{\xi}|}\Bigg|_{\xi_j^-}^{\xi_j^+} d\xi_p d\xi_q - \iiint_V \delta(\boldsymbol{\xi})\frac{\partial}{\partial\xi_j}\left(\frac{1}{|\boldsymbol{x}-\boldsymbol{\xi}|}\right)\mathrm{d}V\right) \\ &= \boldsymbol{e}_i\frac{M_{ij}(t)}{4\pi}\left(-\iiint_V \delta(\boldsymbol{\xi})\frac{x_j-\xi_j}{|\boldsymbol{x}-\boldsymbol{\xi}|^3}\mathrm{d}V\right) \\ &= -\boldsymbol{e}_i\frac{M_{ij}(t)}{4\pi}\frac{x_j}{|\boldsymbol{x}|^3}\end{aligned} \tag{66}$$

The scalar potential:

$$\begin{aligned}\Phi(\boldsymbol{x}) &= \nabla\cdot\mathbf{W}(\boldsymbol{x}) = \frac{\partial}{\partial x_n}\left(-\frac{M_{nj}(t)}{4\pi}\frac{x_j}{|\boldsymbol{x}|^3}\right) \\ &= -\frac{M_{nj}(t)}{4\pi}\left(\frac{\delta_{jn}|\boldsymbol{x}|^3 - 3x_j|\boldsymbol{x}|^2\frac{x_n}{|\boldsymbol{x}|}}{|\boldsymbol{x}|^6}\right) \\ &= -\frac{M_{nj}(t)}{4\pi}\left(\frac{\delta_{jn} - 3x_jx_n/|\boldsymbol{x}|^2}{|\boldsymbol{x}|^3}\right) \\ &= -\frac{1}{4\pi}\left(\frac{M_{nn}}{|\boldsymbol{x}|^3} - \frac{3M_{nj}x_nx_j}{|\boldsymbol{x}|^5}\right)\end{aligned} \tag{67}$$

The vector potential:

$$\begin{aligned}\Psi(\boldsymbol{x}) &= -\nabla\times\mathbf{W}(\boldsymbol{x}) = -\varepsilon_{mnp}\frac{\partial}{\partial x_m}\left(-\frac{M_{nj}(t)}{4\pi}\frac{x_j}{|\boldsymbol{x}|^3}\right)\boldsymbol{e}_p \\ &= \frac{M_{nj}(t)}{4\pi}\varepsilon_{mnp}\frac{\partial}{\partial x_m}\left(\frac{x_j}{|\boldsymbol{x}|^3}\right)\boldsymbol{e}_p \\ &= \frac{M_{nj}(t)}{4\pi}\varepsilon_{mnp}\left(\frac{\delta_{jm}|\boldsymbol{x}|^3 - 3x_j|\boldsymbol{x}|^2\frac{x_m}{|\boldsymbol{x}|}}{|\boldsymbol{x}|^6}\right)\boldsymbol{e}_p \\ &= \frac{M_{nj}(t)}{4\pi}\varepsilon_{mnp}\left(\frac{\delta_{jm} - 3x_jx_m/|\boldsymbol{x}|^2}{|\boldsymbol{x}|^3}\right)\boldsymbol{e}_p \\ &= \left(\frac{\varepsilon_{mnp}M_{nm}(t)}{4\pi|\boldsymbol{x}|^3} - \frac{3}{4\pi|\boldsymbol{x}|^5}\varepsilon_{mnp}M_{nj}(t)x_jx_m\right)\boldsymbol{e}_p\end{aligned} \tag{68}$$

(3) $\mathbf{f}(\boldsymbol{x}) = -M_0(t)\nabla\delta(\boldsymbol{x})$
The generating function:

$$\begin{aligned}
\mathbf{W}(\boldsymbol{x}) &= -\iiint_V \frac{-M_0(t)\nabla_{\boldsymbol{\xi}}\delta(\boldsymbol{\xi})}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|}\mathrm{d}V = \boldsymbol{e}_p\frac{M_0(t)}{4\pi}\iiint_V \frac{1}{|\boldsymbol{x}-\boldsymbol{\xi}|}\frac{\partial\delta(\boldsymbol{\xi})}{\partial\xi_p}d\xi_p d\xi_m d\xi_n \\
&= \boldsymbol{e}_p\frac{M_0(t)}{4\pi}\left(\iint \frac{\delta(\boldsymbol{\xi})}{|\boldsymbol{x}-\boldsymbol{\xi}|}\bigg|_{\xi_p^-}^{\xi_p^+} d\xi_m d\xi_n - \iiint_V \delta(\boldsymbol{\xi})\frac{\partial}{\partial\xi_p}\left(\frac{1}{|\boldsymbol{x}-\boldsymbol{\xi}|}\right)\mathrm{d}V\right) \\
&= -\boldsymbol{e}_p\frac{M_0(t)}{4\pi}\iiint_V \delta(\boldsymbol{\xi})\frac{x_p-\xi_p}{|\boldsymbol{x}-\boldsymbol{\xi}|^3}\mathrm{d}V(\boldsymbol{\xi}) \\
&= -\boldsymbol{e}_p\frac{M_0(t)x_p}{4\pi|\boldsymbol{x}|^3} \\
&= -\frac{M_0(t)}{4\pi}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}
\end{aligned} \tag{69}$$

The scalar potential:

$$\Phi(\boldsymbol{x}) = \nabla\cdot\mathbf{W}(\boldsymbol{x}) = \nabla\cdot\left(-\frac{M_0(t)}{4\pi}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}\right) = \frac{M_0(t)}{4\pi}\nabla\cdot\left(-\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}\right) = \frac{M_0(t)}{4\pi}\nabla^2\left(\frac{1}{|\boldsymbol{x}|}\right) \tag{70}$$

By Problem 3, $\nabla^2\left(-\frac{1}{4\pi|\boldsymbol{x}-\boldsymbol{\xi}|}\right) = \delta(\boldsymbol{x}-\boldsymbol{\xi})$. With $\boldsymbol{\xi}=0$:

$$\nabla^2\left(\frac{1}{|\boldsymbol{x}|}\right) = -4\pi\delta(\boldsymbol{x}) \tag{71}$$

Substituting (71) into (70):

$$\Phi(\boldsymbol{x}) = \frac{M_0(t)}{4\pi}(-4\pi\delta(\boldsymbol{x})) = -M_0(t)\delta(\boldsymbol{x}) \tag{72}$$

The vector potential:

$$\begin{aligned}
\Psi(\boldsymbol{x}) &= -\nabla\times\mathbf{W}(\boldsymbol{x}) = -\varepsilon_{mnp}\frac{\partial}{\partial x_m}\left(-\frac{M_0(t)x_n}{4\pi|\boldsymbol{x}|^3}\right)\boldsymbol{e}_p \\
&= \frac{M_0(t)}{4\pi}\varepsilon_{mnp}\frac{\partial}{\partial x_m}\left(\frac{x_n}{|\boldsymbol{x}|^3}\right)\boldsymbol{e}_p \\
&= \frac{M_0(t)}{4\pi}\varepsilon_{mnp}\left(\frac{\delta_{nm}}{|\boldsymbol{x}|^3} - \frac{3x_nx_m}{|\boldsymbol{x}|^5}\right)\boldsymbol{e}_p \\
&= \boldsymbol{e}_p\frac{M_0(t)}{4\pi}\left(\frac{\varepsilon_{mmp}}{|\boldsymbol{x}|^3} - \frac{3\varepsilon_{mnp}x_nx_m}{|\boldsymbol{x}|^5}\right)\boldsymbol{e}_p
\end{aligned} \tag{73}$$

Since $\varepsilon_{mmp} = 0$ (repeated index) and $\varepsilon_{mnp}x_nx_m = \varepsilon_{mnp}(x_nx_m + x_mx_n)/2 = (\varepsilon_{mnp} + \varepsilon_{nmp})x_nx_m/2 = 0$ by antisymmetry:

$$\Psi(\boldsymbol{x}) = 0 \tag{74}$$